

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Data Exploration

```
In [2]: #Data Load and first visual
ruta_archivo = "../data/benchmark_ai_detection_multimodel_2026.csv"
df = pd.read_csv(ruta_archivo)
df.head()
```

Out[2]:

	id	timestamp	domain_context	source_model	is_ai_generated
0	de00499a-a413-4d81-b07d-2ce1c54dd27b	2025-01-06 15:39:56	Medical_Diagnosis_Summary	GPT-5_Preview	1
1	6b4603c7-21c3-4290-830a-c9bff4cf6842	2025-01-06 18:56:37	Medical_Diagnosis_Summary	Claude-4_Opus	1
2	98e69cdf-4b68-4606-874f-b97a8775d5d7	2025-01-08 13:19:23	Legal_Contract_Review	Gemini-2_Pro	1
3	0d770e65-4657-42da-b571-b71faf062759	2025-01-08 23:50:36	Legal_Contract_Review	GPT-5_Preview	1
4	10b7bbdd-ed0c-4fd0-b183-c2f2735d2687	2025-01-09 11:48:37	Software_Architecture_Doc	Gemini-2_Pro	1

Columns Descriptor

- id: Unique identifier for each text sample.
- timestamp: Date and time of the document creation.
- domain_context: The professional field of the text (e.g., Cybersecurity, Medical).
- source_model: The specific creator of the text (Human or specific AI model).
- is_ai_generated: Binary label (0 = Human, 1 = AI). Target variable for classification.
- text_content: The actual text paragraph to be analyzed.
- word_count: Total number of words in the text.

readability_score: Flesch Reading Ease score (higher = easier to read).
avg_sentence_length: Average number of words per sentence (proxy for burstiness).
simulated_perplexity: A statistical estimation of text randomness (AI tends to have lower perplexity).
label_confidence: The confidence score of the source attribution (simulated).

The first two columns are metadata generated during the creation of the dataset, so we will not take them into consideration.

The target variable we want to predict is "is_ai_generated".

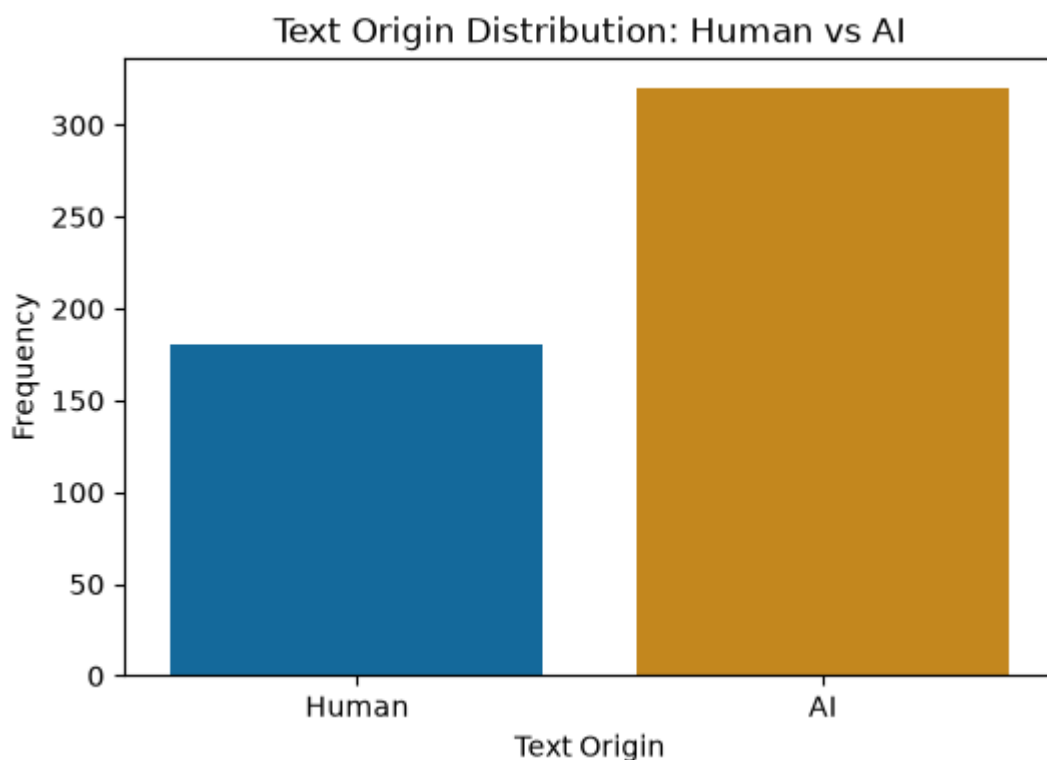
The "source_model" feature should not be used for training, but in this step of the project, could be relevant for exploration.

Visualizations

```
In [3]: plt.figure(figsize=(6, 4))
sns.countplot(data=df, x='is_ai_generated', hue='is_ai_generated', palette='colorblind')

plt.title('Text Origin Distribution: Human vs AI')
plt.xlabel('Text Origin')
plt.ylabel('Frequency')
plt.xticks(ticks=[0, 1], labels=['Human', 'AI'])

plt.show()
```

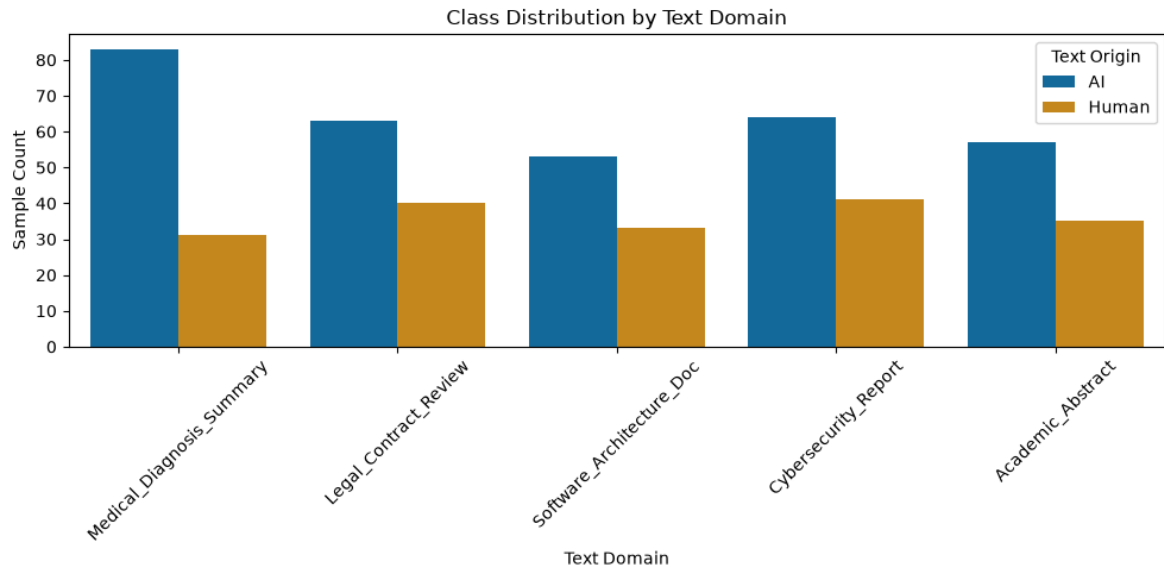


We can see that the Human class is a minority, but it is not disbalanced.

```
In [4]: plt.figure(figsize=(10, 5))
sns.countplot(data=df, x='domain_context',
              hue=df['is_ai_generated'].map({0: 'Human', 1: 'AI'}),
              palette='colorblind')
```

```
plt.title('Class Distribution by Text Domain')
plt.xlabel('Text Domain')
plt.ylabel('Sample Count')
plt.xticks(rotation=45)
plt.legend(title='Text Origin')
plt.tight_layout()

plt.show()
```

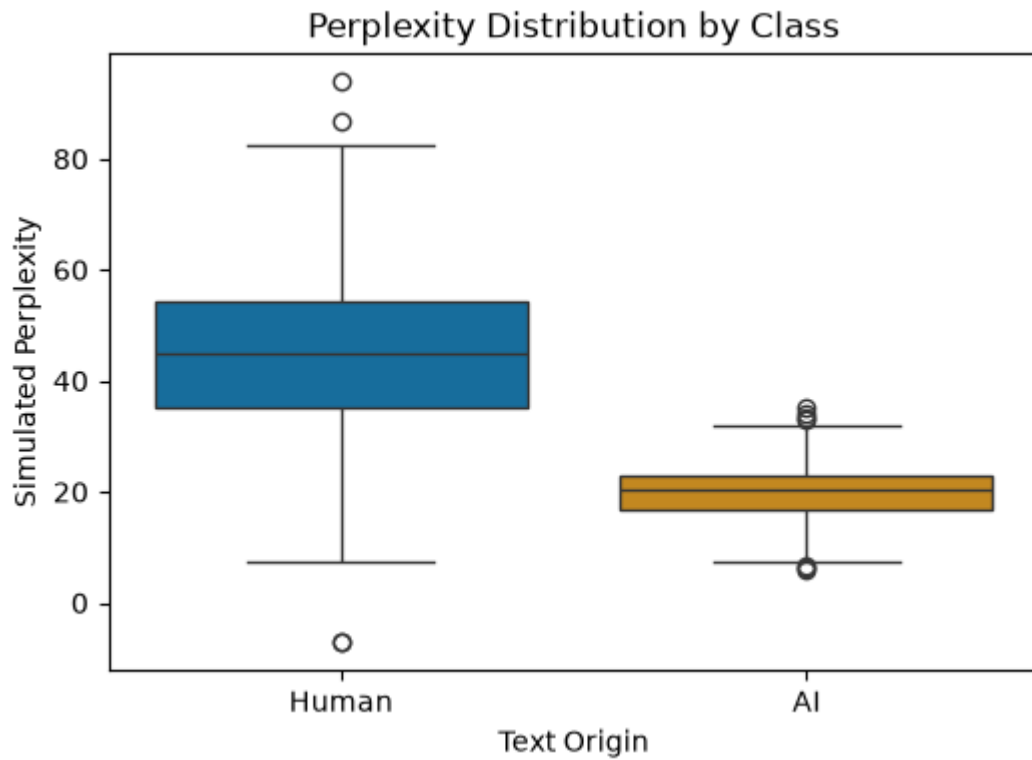


The proportion is similar between the different topics

```
In [5]: plt.figure(figsize=(6, 4))
sns.boxplot(data=df, x='is_ai_generated', y='simulated_perplexity',
            hue='is_ai_generated', palette='colorblind', legend=False)

plt.title('Perplexity Distribution by Class')
plt.xlabel('Text Origin')
plt.ylabel('Simulated Perplexity')
plt.xticks(ticks=[0, 1], labels=['Human', 'AI'])

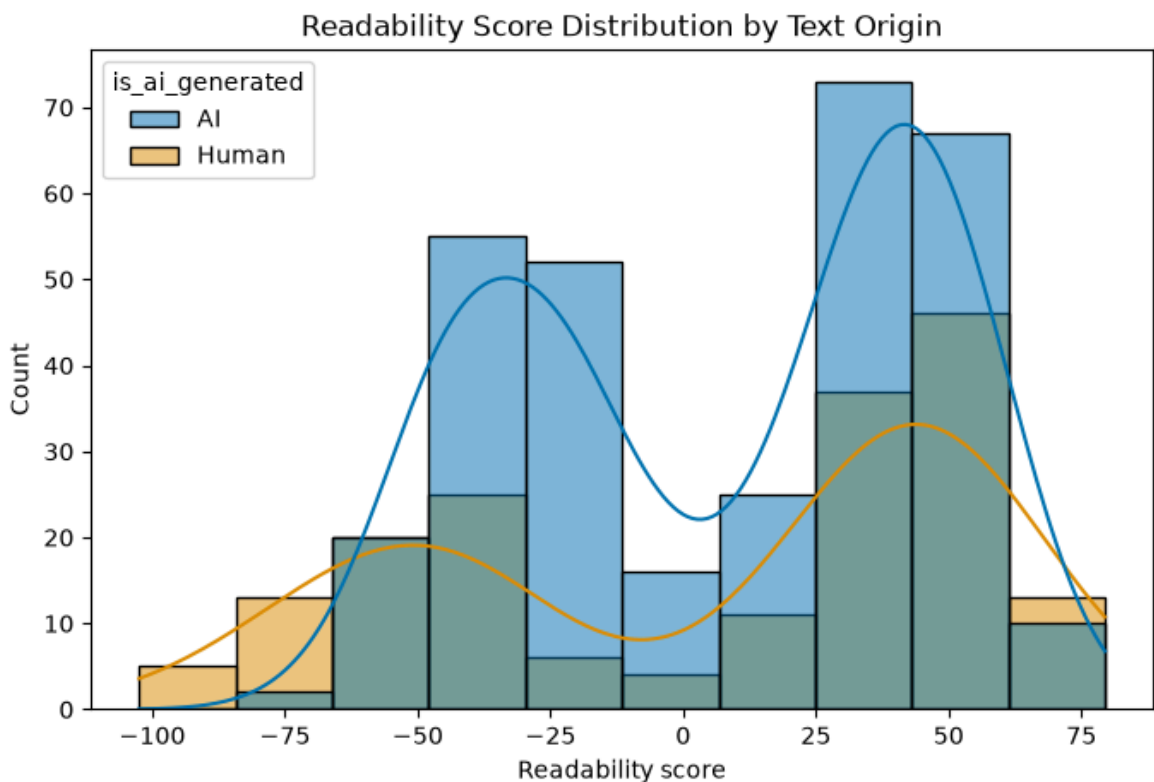
plt.show()
```



```
In [6]: plt.figure(figsize=(8, 5))
sns.histplot(data=df, x='readability_score',
             hue=df['is_ai_generated'].map({0: 'Human', 1: 'AI'}),
             kde=True, palette='colorblind', alpha=0.5)

plt.title('Readability Score Distribution by Text Origin')
plt.xlabel('Readability score')
plt.ylabel('Count')

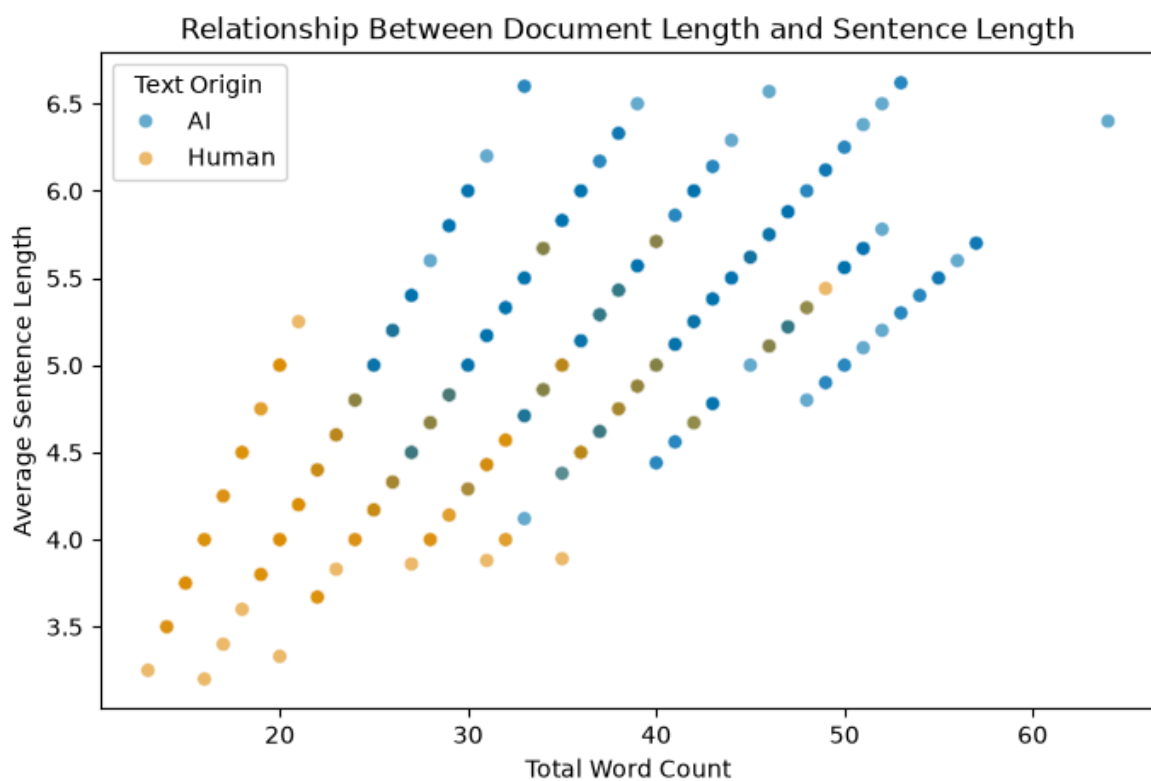
plt.show()
```



```
In [7]: plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x='word_count', y='avg_sentence_length',
                hue=df['is_ai_generated'].map({0: 'Human', 1: 'AI'}),
                alpha=0.6, palette='colorblind')

plt.title('Relationship Between Document Length and Sentence Length')
plt.xlabel('Total Word Count')
plt.ylabel('Average Sentence Length')
plt.legend(title='Text Origin')

plt.show()
```



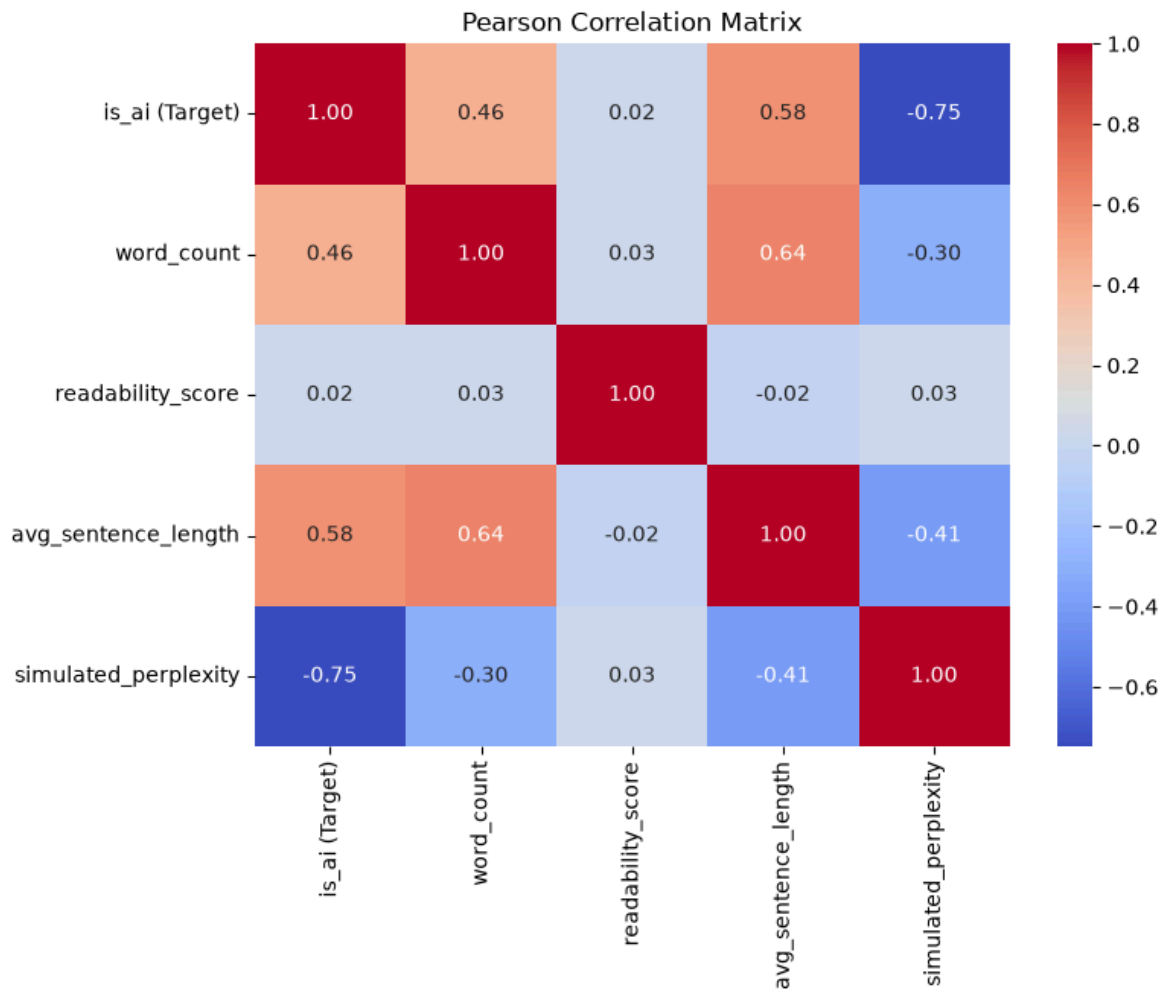
It seems that usually human text tend to be simpler, shorter and less complex

```
In [8]: plt.figure(figsize=(8, 6))
numeric_columns = ['is_ai_generated', 'word_count', 'readability_score',
                  'avg_sentence_length', 'simulated_perplexity']
corr_matrix = df[numeric_columns].corr()

corr_matrix.rename(index={'is_ai_generated': 'is_ai (Target)'},
                  columns={'is_ai_generated': 'is_ai (Target)'}, inplace=True)

sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Pearson Correlation Matrix')

plt.show()
```

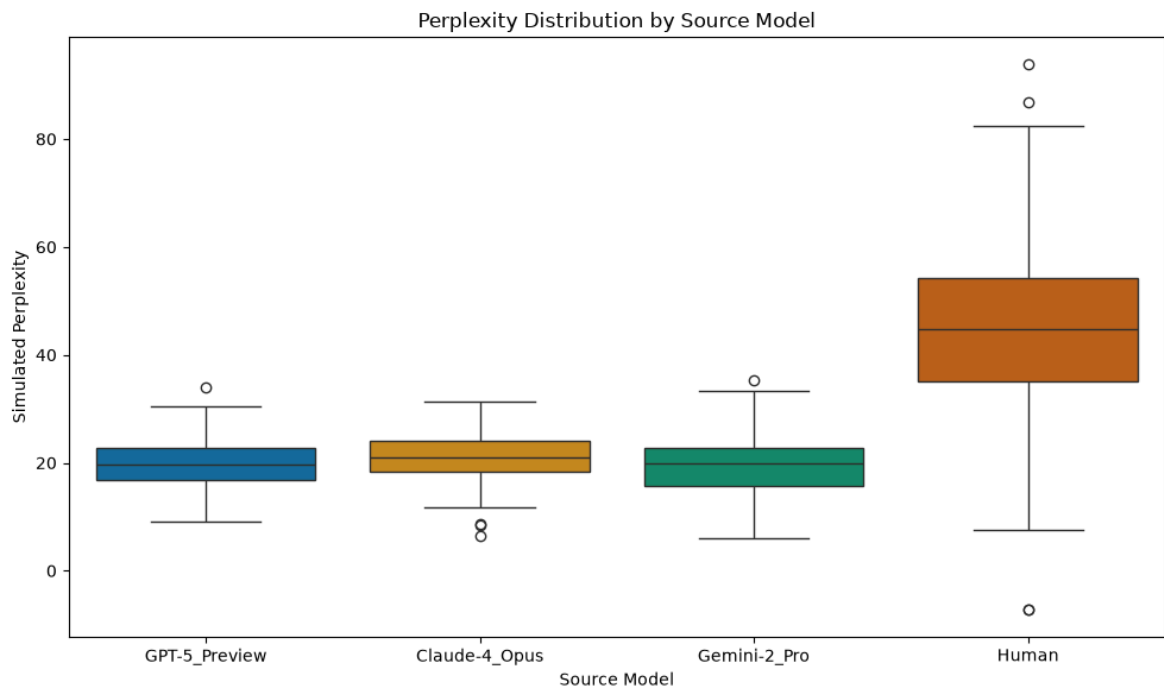


The different ia models have similar behaviour

```
In [9]: plt.figure(figsize=(10, 6))
sns.boxplot(data=df, x='source_model', y='simulated_perplexity',
            hue='source_model', palette='colorblind', legend=False)

plt.title('Perplexity Distribution by Source Model')
plt.xlabel('Source Model')
plt.ylabel('Simulated Perplexity')
plt.tight_layout()

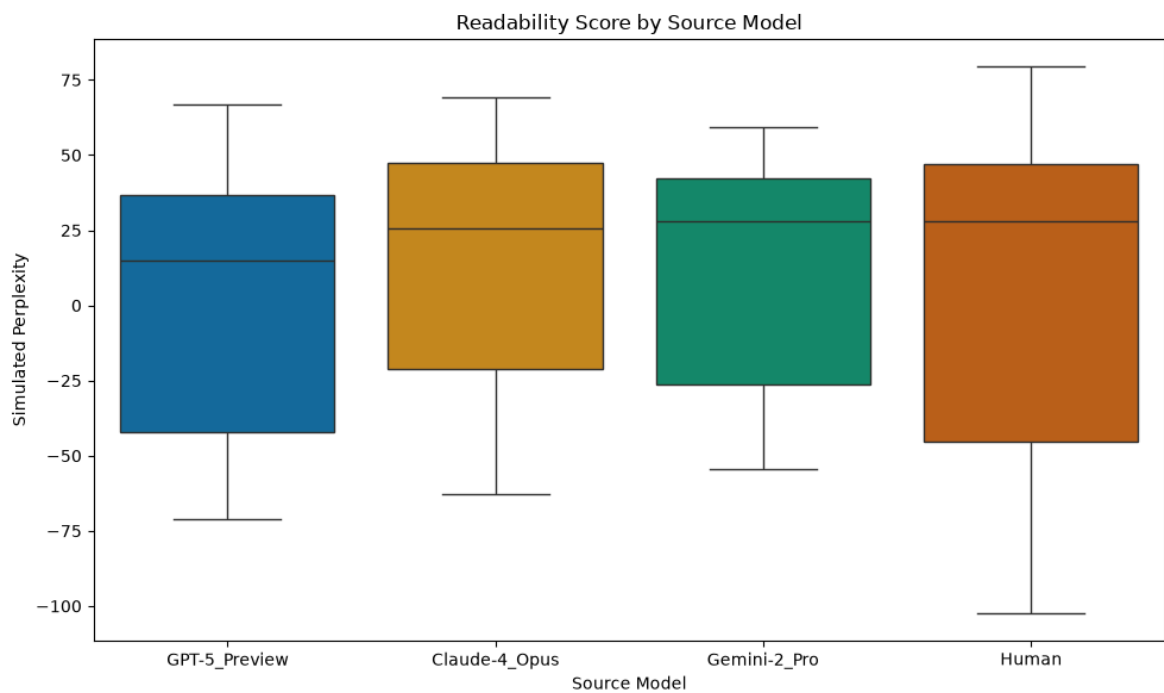
plt.show()
```



```
In [10]: plt.figure(figsize=(10, 6))
sns.boxplot(data=df, x='source_model', y='readability_score',
            hue='source_model', palette='colorblind', legend=False)

plt.title('Readability Score by Source Model')
plt.xlabel('Source Model')
plt.ylabel('Simulated Perplexity')
plt.tight_layout()

plt.show()
```



Resumen

```
In [11]: numeric_columns = ['is_ai_generated', 'word_count', 'readability_score',
                             'avg_sentence_length', 'simulated_perplexity']
```

```

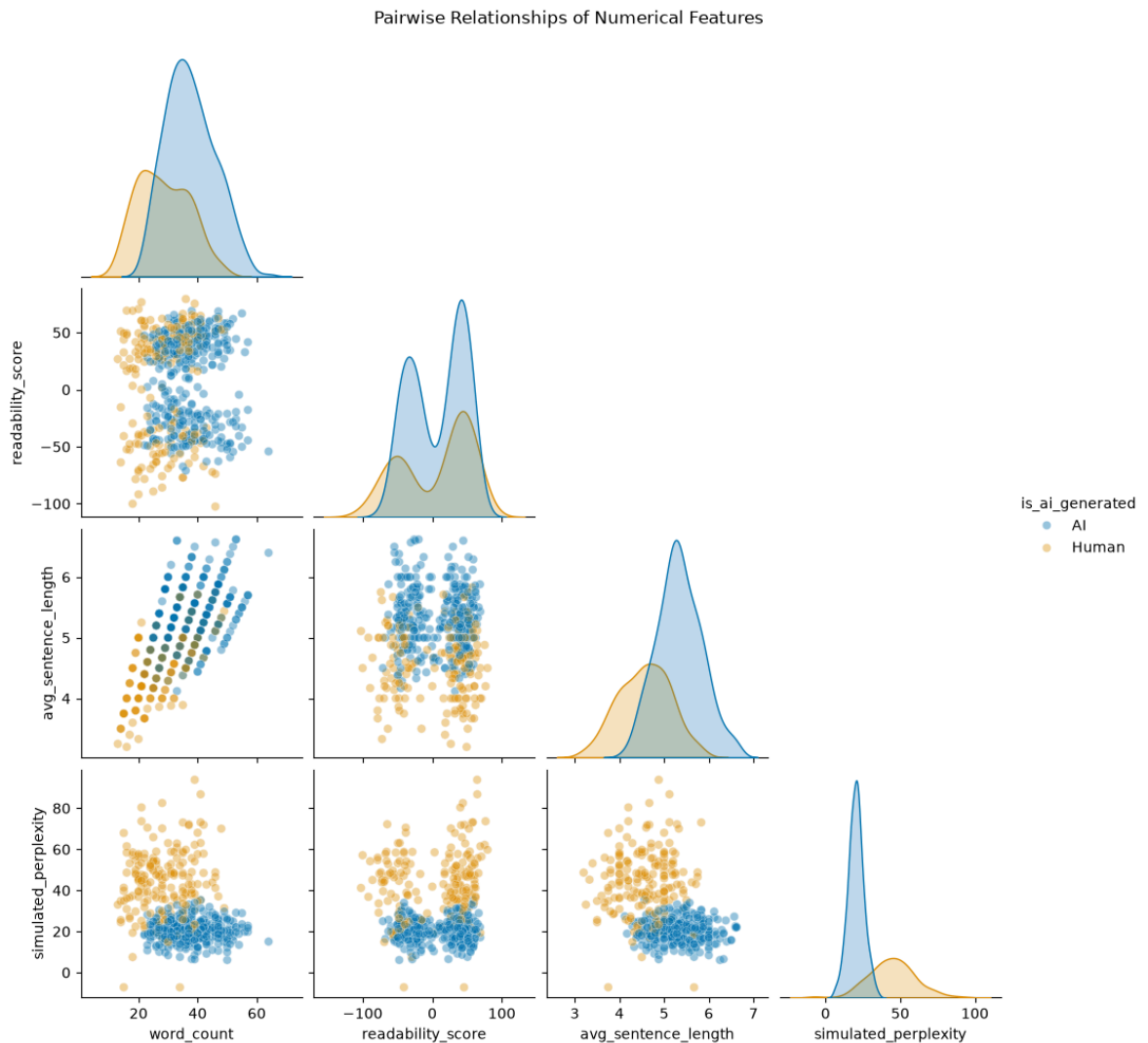
df_pair = df[numeric_columns].copy()
df_pair['is_ai_generated'] = df_pair['is_ai_generated'].map({0: 'Human', 1: 'AI'})

sns.pairplot(df_pair, hue='is_ai_generated', palette='colorblind', corner=True, p

plt.suptitle('Pairwise Relationships of Numerical Features', y=1.02)

plt.show()

```



```

In [12]: plt.figure(figsize=(10, 8))

columnas_pairplot = ['source_model', 'word_count', 'readability_score',
                    'avg_sentence_length', 'simulated_perplexity']

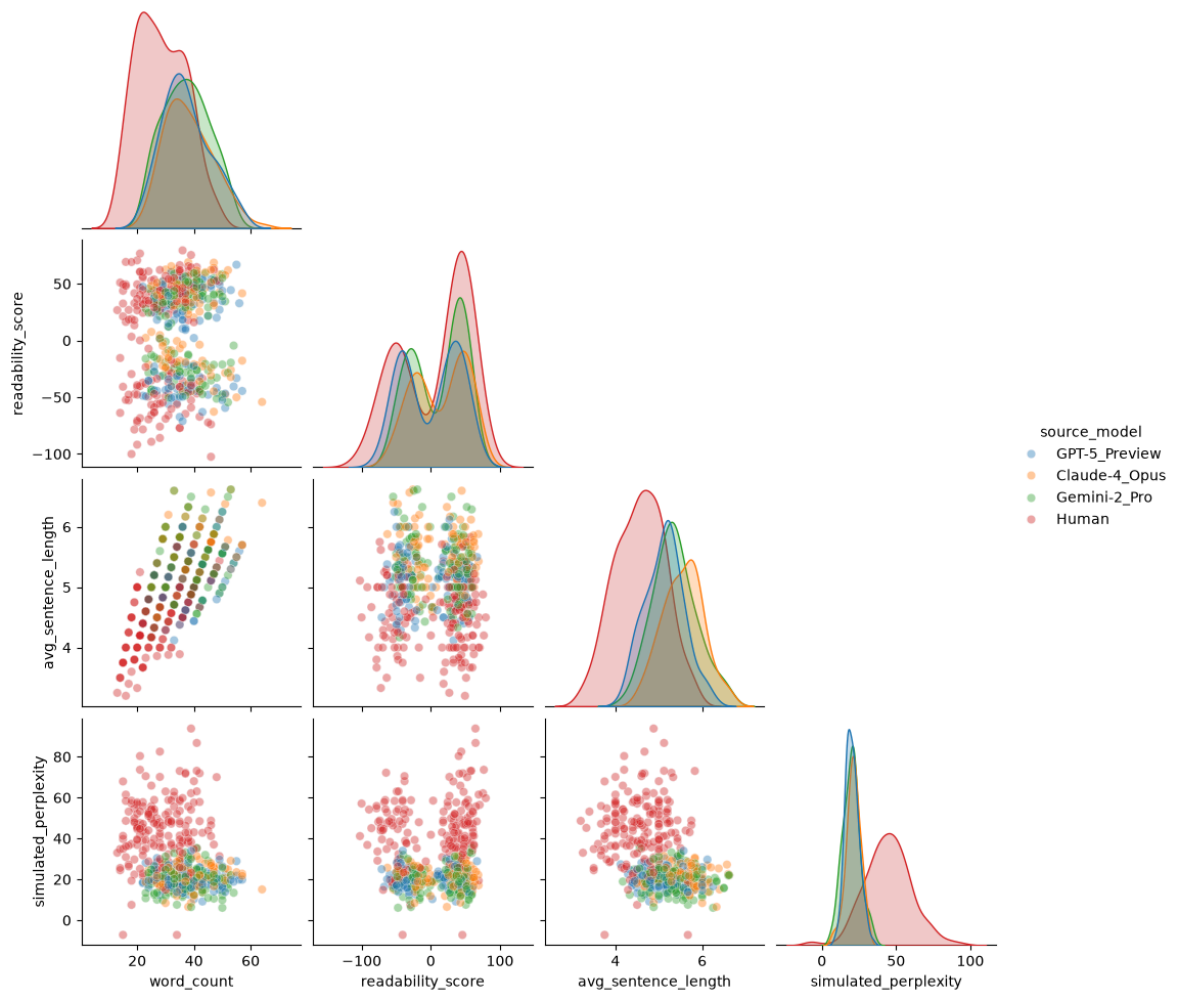
df_pair_model = df[columnas_pairplot].copy()
sns.pairplot(df_pair_model, hue='source_model', palette='tab10', corner=True, pl
plt.suptitle('Pairwise Relationships by Source Model', y=1.02)

plt.show()

```

<Figure size 1000x800 with 0 Axes>

Pairwise Relationships by Source Model



In the graphs of the numeric variables we can see that both classes should be easily separable